

Cardiac Phase, Genomic Profile, and Emotional Memory Consolidation:
*The Moderating Roles of BDNF Val66Met and ADRA2B Deletion Variants on
Systolic Interference*

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Abstract

In earlier experiments, it has been discovered that different stages of cardiac function impact the recollection of non-emotional stimuli in individuals. Specifically, research by Garfinkel et al. (2013) demonstrated that non-emotional stimuli are better remembered during the diastolic phase than during the systolic phase. However, it is not certain if a similar relationship exists between cardiac phases and the recollection of emotional stimuli, or how individual differences in genomics moderate this interaction. Since it cannot be deduced for certain if cardiac phases impact emotional memory as they do non-emotional memory, we propose research to determine the correlation between the recollection of valenced stimuli, cardiac function, and specific genetic polymorphisms. This experiment seeks to determine if the relationship between emotional memory and cardiac function is moderated by variations in the BDNF Val66Met and ADRA2B deletion variants. Ultimately, we aim to discover if specific genomic profiles protect against or amplify the "systolic interference" effect on memory.

Introduction

The integration of physiological arousal with cognitive processing is a cornerstone of affective science, tracing back to the James-Lange theory, which proposes that bodily sensations provide the experiential quality of emotion. Central to this integration is interoception — the process by which the brain senses, interprets, and integrates signals originating from within the body (Garfinkel et al., 2013). Unlike the exteroceptive senses, which orient an individual toward the external world, interoception governs awareness of one's internal physiological state, encompassing pain, hunger, arousal, and emotion (Price & Hooven, 2018). It is through this interoceptive channel that one major source of visceral information reaches higher cortical regions: baroreceptor firing, which occurs in bursts during cardiac systole and remains silent during diastole. While these afferent signals are critical for homeostatic regulation, they also impinge

upon cognitive functions, acting as a form of "visceral interference" that can disrupt the encoding of information during systole (Garfinkel et al., 2013).

Emotional processing is particularly vulnerable to such interference, given its heavy reliance on interoceptive awareness. The recognition and regulation of emotion requires accurate detection of internal physiological changes, and disruptions to this sensitivity — whether through heightened or blunted awareness — are associated with a range of psychiatric conditions including anxiety, depression, PTSD, and substance use disorder (Khoury et al., 2018). Critically, interoceptive sensitivity varies meaningfully across individuals: anxious individuals tend to exhibit heightened responsiveness to internal signals (Barker et al., 2021), while those with depression demonstrate markedly blunted awareness, consistent with elevated alexithymia (Murphy et al., 2018). This individual variability is directly relevant to the present proposal, as it suggests that the degree of systolic interference on emotional memory encoding will not be uniform across participants.

At the neural level, interoceptive signals are integrated primarily within the insula and the anterior cingulate cortex — regions critically involved in emotional appraisal and memory consolidation (Garfinkel et al., 2013; Price & Hooven, 2018). Emerging evidence indicates that individual differences in these circuits are substantially moderated by genomic variation. The ADRA2B deletion variant is associated with heightened amygdala activation during the encoding of negative emotional stimuli and stronger functional connectivity between the amygdala and insula — the very pathway through which interoceptive signals reach emotional processing centers (Rasch et al., 2009). Similarly, the BDNF Val66Met polymorphism influences synaptic plasticity and neural reactivity in the amygdala and hippocampus, with Met carriers exhibiting greater activation in response to emotional stimuli than Val homozygotes (Cathomas et al., 2010; Nicholson et al., 2023). These genetic profiles offer a plausible biological mechanism for why individuals differ in how intensely they encode emotionally valenced interoceptive signals and, by extension, how susceptible they may be to systolic interference during emotional memory formation. This proposal aims to bridge autonomic physiology and genomics to determine whether specific genetic variants protect against or amplify the systolic interference effect on emotional memory.

Memory: Emotional, Explicit, and the Role of Physiological and Genomic Modulation

Memory is broadly divisible into emotional and non-emotional types, a distinction with significant neuroanatomical grounding. Non-emotional memory encodes informational content that does not substantially alter an individual's affective state, whereas emotional memory pertains to events that evoke intense arousal and feeling — and is, as a consequence, typically consolidated with greater fidelity (American Psychological Association, 2022). Within emotional memory, a further distinction exists between implicit and explicit forms. Implicit emotional memories are encoded unconsciously, with the amygdala serving as the primary substrate, relying on adrenergic and dopaminergic mechanisms in the subcortical limbic system. Explicit emotional memories, by contrast, are consciously accessible and manifest when individuals re-experience the emotions originally generated by an event; their encoding depends critically on cortical limbic structures, particularly the hippocampus (American Psychological Association, 2022). It is within this explicit, episodic dimension of emotional memory — encompassing autobiographical experiences tied to specific affective states — that the present proposal is principally concerned.

The genomic variants introduced above exert meaningful influence over these memory systems. The BDNF Val66Met polymorphism plays a dual role: while Met carriers tend to exhibit reduced episodic memory for neutral stimuli (Cathomas et al., 2010), this same variant interacts with overnight consolidation to preferentially enhance the retention of emotional images, suggesting a selective advantage in affective memory stabilization (Harrington et al., 2019). The ADRA2B deletion variant similarly augments emotional memory, operating through increased noradrenergic availability during voluntary retrieval of emotionally valenced experiences (Rasch et al., 2009). Notably, both variants exert their effects through the amygdala and hippocampus — the same regions identified above as central to interoceptive signal integration — positioning them as key moderators of how physiological arousal during cardiac systole is translated into lasting emotional memory.

Physiologically, cardiac systole has been demonstrated to suppress memory encoding, consistent with the visceral interference framework outlined in the preceding section (Garfinkel et al., 2013). However, this suppression is not uniform: individuals with higher interoceptive sensitivity, who maintain more precise predictive models of their internal states, appear better equipped to attenuate the disruptive effect of systolic baroreceptor firing on encoding. Taken together, the interplay between genomic profile, interoceptive sensitivity, and cardiac phase creates a multidimensional

framework within which individual differences in emotional memory consolidation can be understood — and which this proposal directly seeks to characterize.

Epigenetic Mechanisms in Emotional Memory Consolidation

Beyond genomic polymorphisms, epigenetic regulation provides an additional layer through which environmental exposures shape emotional memory consolidation. Activity-dependent modifications to chromatin structure allow experiences such as trauma, stress, and sleep to produce lasting changes in the transcription of memory-related genes. In particular, learning-induced alterations in DNA methylation at BDNF promoters have been causally linked to fear memory consolidation, with experimental manipulation of methylation patterns shown to disrupt exon-specific BDNF transcription and impair memory formation (Lubin et al., 2008;). Complementing these methylation dynamics, histone modifications are rapidly and selectively regulated following fear conditioning: increases in H3K4me3 at promoters of immediate-early genes and experience-dependent shifts in H3K9me2 have been observed at loci including BDNF and ZIF268, facilitating the transcriptional programs required for long-term consolidation (Gupta et al., 2010; Hunter et al., 2009; Jarome & Lubin, 2018;). Together, these chromatin-level mechanisms provide a biologically plausible route through which environmental history interacts with genomic predisposition—offering a framework for understanding how prior stress or trauma may modulate the functional impact of variants such as BDNF Val66Met during emotional memory formation.

Interoception and Cardiac Activity

Cardiac activity serves as a primary physiological index of interoceptive awareness, given that blood pressure and cardiac timing are known to influence perceptual, cognitive, and emotional processing (Garfinkel et al., 2013). The mechanism underlying this influence centers on arterial baroreceptors, which fire in bursts during systole as the heart contracts and remain silent during diastole. These afferent signals, while essential for homeostatic regulation, simultaneously compete for neural resources involved in encoding — producing what has been characterized as "visceral interference." Garfinkel et al. (2013) provided the foundational empirical demonstration of this effect: participants presented with non-emotional word stimuli during systole recalled them

significantly less accurately than those presented during diastole, establishing a clear cardiac phase-dependent modulation of memory encoding.

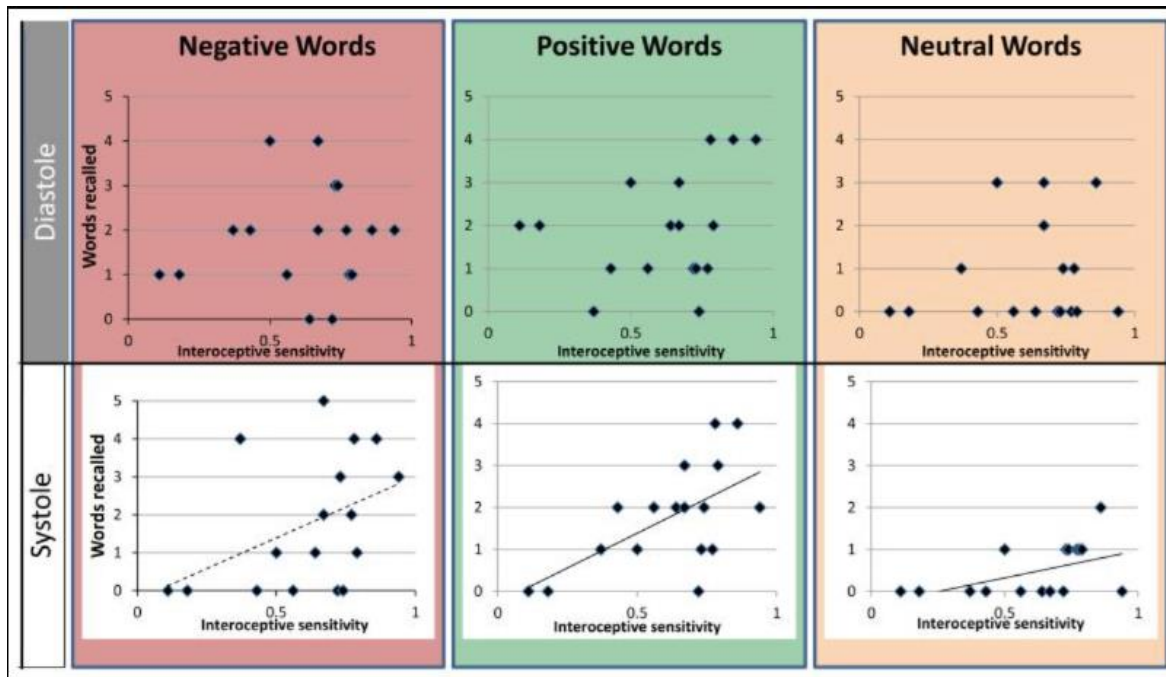


Figure 1
“Graphs depicting words originally detected after presentation on diastole are shown in the upper half, with words originally detected after presentation on systole presented in the lower half” (Garfinkel et al., 2013).

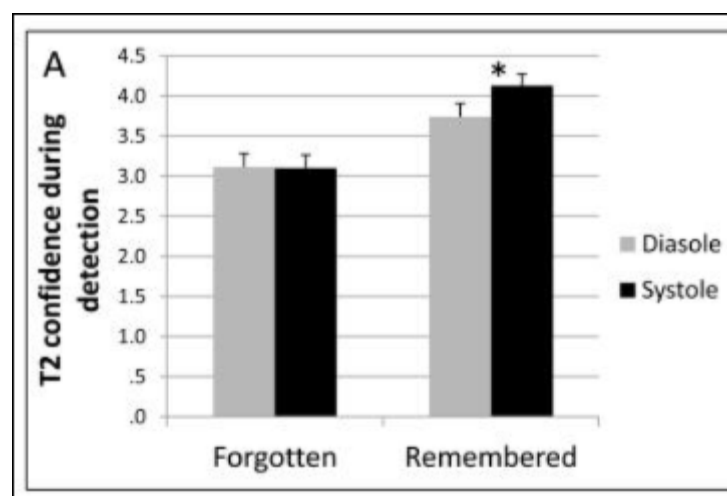


Figure 2

“Memory for words processed at systole tended to encompass only those detected with high confidence, whereas words at diastole that were subsequently recalled were associated with both high and low confidence items as reflected in an overall lower mean” (Garfinkel et al., 2013).

The present proposal extends this framework to emotional memory. Given that emotional stimuli generate heightened bodily arousal and recruit the amygdala and hippocampus with greater intensity than neutral stimuli, it remains an open question whether this arousal can override systolic interference — or whether the interference is similarly, or even differentially, disruptive for emotionally valenced material. We hypothesize that emotional stimuli will be better recalled overall than non-emotional stimuli, consistent with Payne et al. (2008), who demonstrated that negatively arousing objects are more robustly retained and that sleep selectively consolidates emotional — but not neutral — memory. To test this, our experiment will present participants with 30 stimuli: 15 negatively arousing images and 15 photographs of everyday inanimate objects, both in black and white, timed to cardiac phase using baroreceptor recording — an approach modeled directly on Garfinkel et al. (2013).

Critically, however, this proposal moves beyond a simple replication by incorporating genomic stratification. As established in the preceding sections, the ADRA2B deletion variant and the BDNF Val66Met polymorphism each modulate the neural sensitivity of the amygdala and hippocampus to emotionally valenced signals (Rasch et al., 2009; Harrington et al., 2019; Cathomas et al., 2010). By integrating participants' genomic profiles with their cardiac phase data and memory performance, this experiment directly tests whether these variants alter the brain's vulnerability to systolic interference — and whether specific genomic profiles amplify or protect against its disruptive effect on emotional memory consolidation.

Method

Participants

One hundred twenty healthy volunteers will be recruited from the northwestern United States, with geographic specificity chosen to minimize the influence of unknown environmental variables on memory performance. Participants will range in age from 18 to 45 years, providing sufficient demographic breadth to avoid results specific to any single age cohort's memory profile. The sample will be divided equally between males and females (60 per group) to control for potential

gender-related differences in emotional memory or interoceptive sensitivity (Deng et al., 2016). All participants will be screened to exclude individuals with a history of psychiatric, neurological, or sleep disorders, as well as any cardiac conditions that substantially alter systolic or diastolic activity or produce irregular ECG readings, which would confound cardiac phase timing. Saliva samples will be collected using standard kits to genotype each participant for the BDNF Val66Met and ADRA2B deletion variants (Cathomas et al., 2010; Rasch et al., 2009), enabling stratification of the sample by genomic profile. Participants will be instructed to obtain a full night of sleep the evening prior to testing to eliminate sleep deprivation as a confounding influence on encoding.

Stimuli and Apparatus

Stimuli will consist of 60 photographs in total, all standardized in black and white and uniform in size (11×13 inches). Thirty photographs will serve as the primary stimulus set presented during the learning phase: 15 negatively arousing emotional images selected from the International Affective Picture System (IAPS; e.g., depictions of death, atrocities, and war zones) and 15 neutral images depicting common everyday inanimate objects (e.g., a pepper shaker, tissue box, air conditioner). The remaining 30 photographs — equally divided between emotional and non-emotional — will serve as foils during the recognition test and will not be shown to participants prior to that phase. Cardiac activity will be recorded continuously throughout the learning phase using a three-lead electrocardiogram (ECG) and baroreceptor measurements, enabling precise identification of systolic and diastolic phases and detection of cardiac arousal responses to individual stimuli (Garfinkel et al., 2013).



Figure 3.0

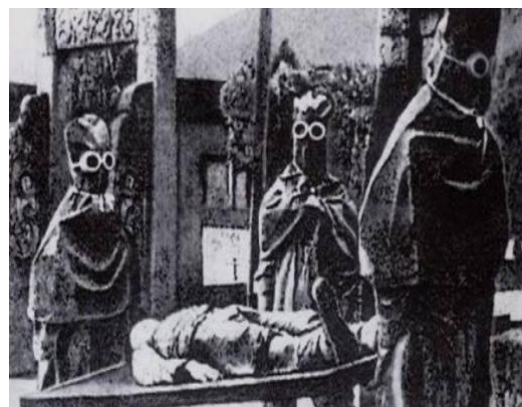


Figure 3.1

These photographs taken in concentration camps during WWII serve as an example of the type of emotional stimuli that will be used in this experiment (Taylor, 2011).



Figure 4.0



Figure 4.1

These photographs of everyday items serve as an example of the type of non-emotional stimuli that will be used in this experiment (iStock, 2008; Holly, 2008).

Procedure

The experiment will follow a modified Rapid Serial Visual Presentation (RSVP) paradigm adapted from Garfinkel et al. (2013). During the learning phase, all 30 primary photographs will be presented in randomized order at a rapid rate of 100ms per image. Each stimulus will be time-locked to either the R-wave (diastole) or T-wave (systole) of the participant's ECG, counterbalanced across participants, such that emotional and non-emotional images are systematically presented across both cardiac phases. Continuous ECG and baroreceptor recordings will be collected throughout this phase to document cardiac timing and identify any significant physiological arousal responses elicited by individual stimuli.

Following the learning phase, a 24-hour delay will be imposed prior to memory testing. This interval incorporates a full night of sleep, which is critical given evidence that sleep selectively consolidates emotional memory and interacts specifically with the BDNF Val66Met polymorphism during overnight retention (Harrington et al., 2019; Payne et al., 2008). At the end of this interval, participants will complete a surprise recognition test in which they are presented with all 60 photographs — the 30 original stimuli intermixed with 30 foils — and asked to identify which images they previously viewed. Participants will be given a maximum of 10 minutes to complete the task (Garfinkel et al., 2013). Memory performance will be analyzed as a function of

stimulus valence, cardiac phase at encoding, and genomic group, enabling direct assessment of whether the BDNF Val66Met and ADRA2B deletion variants moderate susceptibility to systolic interference during emotional memory consolidation.

Epigenetic Assays

The experiment will incorporate epigenetic profiling at both the encoding session and the 24-hour recognition test, enabling assessment of dynamic molecular changes associated with emotional memory consolidation. Saliva or blood samples will be collected at both time points, followed by bisulfite conversion and targeted pyrosequencing of CpG sites within BDNF promoter IV, a region in which learning-induced methylation changes have been causally linked to fear memory consolidation (Lubin et al., 2008;). Additional candidate loci will include FKBP5 and NR3C1, both of which exhibit trauma-associated methylation shifts and predict stress-related memory outcomes and PTSD symptom trajectories (Zannas et al., 2015;). These assays will allow quantification of environmentally responsive methylation patterns that may mediate or moderate the relationship between genotype, cardiac phase, and emotional memory performance.

As an exploratory extension, and where feasible (e.g., in animal follow-up studies or via peripheral chromatin proxies), histone modification profiling will be conducted using ChIP-qPCR. Specifically, enrichment of H3K4me3 and H3K9me2 at *BDNF* and immediate-early gene promoters will be quantified relative to input controls. These marks undergo rapid, experience-dependent regulation following fear conditioning and facilitate transcriptional programs required for long-term consolidation (Gupta et al., 2010;). Reporting fold-enrichment values will provide an additional chromatin-level index of consolidation-related plasticity.

Analysis Plan

Epigenetic variables will be incorporated into the statistical model to determine whether methylation or histone mark levels mediate or moderate the genotype $\times$ cardiac phase $\rightarrow$ memory relationship, using moderated mediation frameworks with bootstrap confidence intervals. This analytic approach enables direct testing of whether epigenetic plasticity serves as a mechanistic bridge between physiological state at encoding and long-term emotional memory retention.

Additional Candidate Loci

To extend the analysis beyond BDNF and ADRA2B, several additional loci implicated in stress-axis and monoaminergic regulation will be examined. FKBP5 and NR3C1, key regulators of the HPA axis, show robust trauma-linked methylation changes and predict long-term trajectories of stress reactivity and memory (Zannas et al., 2015;). SLC6A4 methylation has been associated with altered limbic responses to emotional stimuli (Nikolaidis et al., 2023;), while COMT epigenetic variation has been linked to dopaminergic modulation of cognition and emotional processing (Ursini et al., 2011;). Together, these loci broaden the scope of the study to encompass stress-responsive and monoaminergic pathways implicated in the over-consolidation of traumatic memories.

Hypothesized Results

We anticipate that non-emotional stimuli will replicate the pattern established by Garfinkel et al. (2013), with significantly better recall for images presented during diastole than during systole, confirming that visceral interference disrupts the encoding of neutral material. For emotional stimuli, however, we hypothesize that the heightened bodily arousal generated by negatively valenced images will attenuate or override this systolic suppression, producing no significant difference in recall across cardiac phases — consistent with Payne et al.'s (2008) finding that negatively arousing stimuli are more robustly retained than neutral ones.

With respect to genomic moderation, we predict that ADRA2B deletion carriers and BDNF Met carriers will demonstrate the greatest emotional memory advantage, exhibiting enhanced retention of negatively arousing stimuli regardless of cardiac phase at encoding (Rasch et al., 2009; Harrington et al., 2019). This would implicate elevated noradrenergic availability and heightened amygdala-hippocampal reactivity as mechanisms that buffer against systolic interference specifically for emotionally valenced material (Rasch et al., 2009; Nicholson et al., 2023). Conversely, participants without these variants are expected to show greater susceptibility to cardiac phase effects even for emotional stimuli. Additionally, individuals with higher interoceptive sensitivity, as measured by heartbeat detection performance, are hypothesized to be more resistant to the deleterious effects of systole on encoding (Garfinkel et al., 2013), consistent with the proposal that precise predictive models of internal states reduce the cognitive cost of baroreceptor discharge.

Discussion

If the hypothesized results are confirmed, this experiment would provide the first empirical evidence that the systolic interference effect on memory is valence-dependent — that is, that the brain selectively prioritizes the encoding of threatening or emotionally salient stimuli even during periods of peak baroreceptor discharge. This would constitute a meaningful extension of the Garfinkel et al. (2013) framework, demonstrating that the cardiac phase-memory relationship established for neutral stimuli does not uniformly generalize to emotional material.

The genomic findings carry theoretical and clinical significance. Evidence that ADRA2B and BDNF variants amplify emotional memory consolidation independent of cardiac timing (Rasch et al., 2009; Harrington et al., 2019) would offer a neurobiological account of why certain individuals are disproportionately vulnerable to the over-consolidation of traumatic memories — a hallmark feature of PTSD and related trauma-spectrum disorders (Nicholson et al., 2023). Patterns of BDNF promoter methylation and histone modifications (H3K4me3, H3K9me2) offer a dynamic readout of how prior stress, trauma, or environmental exposures shape the functional expression of the Val66Met polymorphism. Rather than serving as static markers, these chromatin signatures may index whether an individual's molecular landscape biases them toward heightened consolidation or confers resilience against over-encoding emotionally salient events. By identifying the genomic profiles that either protect against or exacerbate this effect, this research could contribute to precision-based approaches to understanding trauma vulnerability, informing both risk stratification and the development of targeted interventions.

The role of interoceptive sensitivity in moderating systolic interference (Garfinkel et al., 2013) further suggests that somatic awareness is not merely a passive correlate of emotional processing, but an active determinant of memory fidelity. This opens a translational avenue for interoception-focused therapies — such as mindfulness-based or biofeedback interventions — as tools for modulating the encoding of emotionally distressing experiences (Price & Hooven, 2018; Khoury et al., 2018).

Several limitations warrant acknowledgment. The 24-hour retention interval, while necessary for capturing sleep-dependent consolidation (Payne et al., 2008), introduces variables that cannot be fully controlled, including the quality and duration of sleep between sessions. Similarly, while

participants will be instructed to rest adequately prior to the learning phase, adherence cannot be verified, and sleep deprivation remains a potential confound. Experimental conditions will be standardized to the greatest extent possible — including uniform image size, quality, and presentation order randomization — to minimize perceptual and attentional biases unrelated to cardiac phase or genomic profile. Future work should consider polysomnographic monitoring to more precisely characterize the sleep-dependent consolidation component (Harrington et al., 2019) and extend the genomic analysis to additional variants implicated in noradrenergic and serotonergic regulation of emotional memory.

Future Directions

The present proposal establishes a foundation for a broader research program examining how genomic profile, cardiac physiology, and interoceptive sensitivity converge to shape emotional memory. Several extensions of this framework merit direct investigation.

A natural first extension concerns gender differences in emotional memory consolidation. While the equal male-female sample in this study is designed to control for gender as a variable, the findings of Deng et al. (2016) suggest it warrants dedicated examination. That study distinguished between emotional experience — which is governed by interoceptive awareness — and emotional expressivity, finding that men exhibited stronger emotional experiences while women demonstrated stronger emotional expressivity. Given that interoceptive experience, not expressivity, is the mechanism through which cardiac phase influences encoding (Garfinkel et al., 2013), and given that women reported higher physiological arousal across all emotion types (Deng et al., 2016), a follow-up experiment directly comparing male and female participants on cardiac phase-dependent emotional memory performance is warranted. We hypothesize that females would demonstrate enhanced retention of negatively arousing stimuli relative to males, mediated by greater arousal-driven consolidation — though this remains to be tested empirically.

Beyond gender, future work should extend this framework to clinical populations, particularly individuals with PTSD or Major Depressive Disorder, in whom the relationship between interoceptive sensitivity and emotional memory is known to be dysregulated (Khoury et al., 2018). Testing whether ADRA2B and BDNF genotypes predict the severity of intrusive memory symptoms or re-experiencing in these populations (Nicholson et al., 2023) would establish a direct

translational link between the mechanisms identified here and real-world psychiatric vulnerability. Additionally, epigenetic factors — such as DNA methylation patterns at the BDNF gene locus — could illuminate how prior trauma exposure modifies the functional impact of the Met allele on memory consolidation (Harrington et al., 2019), providing a mechanism by which environmental history interacts with genomic predisposition. Finally, integrating standardized interoceptive profiling across participants would allow future studies to use interoceptive sensitivity as a continuous predictive biomarker (Garfinkel et al., 2013; Price & Hooven, 2018), rather than a secondary measure, potentially identifying individuals at heightened risk for maladaptive emotional memory formation across a range of psychiatric conditions (Khoury et al., 2018).

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